

# AP-E7: A Common Quadratic ( $Sp(4)$ ) APS/PV Regulator Exact Pure Heavy-Threshold Phase, Two Target Tests, and a Determinant-Line No-Go

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## Abstract

This note repairs a precise omission of AP-E6. For the candidate simply connected group, denoted  $Sp(4)$  in the physics convention and  $Sp(2)$  in the mathematics convention, we specify one common background-field quadratic regulator for fermions, gauge fields, the adjoint and fundamental scalars, and Faddeev–Popov ghosts. The Hilbert spaces, Sobolev domains, Euclidean Clifford and reference real structures, gauge complex, Pauli–Villars superdeterminants, finite local subtraction, five-dimensional mass interpolation, spectral cut, and APS domains are all explicit. This closes `common_quadratic_aps_pv_regulator_specified`. It does not construct a nonperturbative continuum gauge measure: Gribov/global gauge slicing, interacting removal of the cutoff, and reflection positivity remain open.

As a constructive negative control, we also give a compact-link overlap framework and a continuum Higgs-orbit source that are mutually Ginsparg–Wilson compatible. We do *not* write the finite-lattice Dirac–Yukawa operator or its five-dimensional target kernel, so this is not a completed overlap lift and no overlap eta pair is claimed as computed. A copy-counting diagnostic predicts the conditional pair  $(+1, +1)$  if such a lift, the standard determinant orientation, and the no-stacked-torsion convention are supplied. This is new conditional Higgsed-sector data, not a consequence of the original action. A Schur-lemma theorem proves only that an exact separated rank-two light projector cannot extend continuously and  $Sp(4)$ -equivariantly to the restored point  $\Sigma = 0$ .

The same regulator gives a new exact threshold result. For one flavour the negative-mass heavy bundle after  $Sp(4) \rightarrow SU(2)_c \times U(1)_H$  is  $E_h = (V_c \otimes L^{-1}) \oplus \mathbf{1}$ , and

$$I_h = \text{ind } D_{E_h}^+ = \int_{M_4} \left[ -c_2(V_c) + x^2 - \frac{1}{8}p_1(TM_4) \right], \quad x = c_1(L).$$

The two Dirac flavours therefore contribute  $\exp(2\pi i I_h) = 1$  on every closed spin four-manifold. Thus the constant-mass pure unbroken-gauge plus gravitational APS threshold is computed and is removable by the displayed integral local counterterm. On the two target representatives  $G_3 = S_R^1 \times S^3$  and  $G_2 = T_{RR}^2 \times S^2$ , the same pure component is explicitly  $(+1, +1)$ .

This does not determine the full target phases. The microscopic action has no map from  $(g, s) \in S^3 \times S^2$  to a full all-branch chiral Dirac–Yukawa/PV family:  $g$  is composite and no  $s$ -dependent fermion mass is declared. Consequently there is no pullback determinant line over  $\text{Map}(M_4, S^3 \times S^2)$ . We prove non-identifiability by stacking the two independent zero-curvature spin characters  $(-1)^{\nu_3}$  and  $(-1)^{\nu_2}$ , which leaves every local symbol and the pure threshold unchanged while flipping either target sign. Moreover, two controlled ansatz have been evaluated: a uniform exact flavour mass map on the complete  $\mathbf{5}$  has mixed coefficient  $+2 - 2 = 0$ , whereas a light-only emergent flavour assignment has no declared deep-UV target family. These cases do not exhaust continuous equivariant non-projector couplings. Hence the present work does not give an all-scale  $k = +2$  matching theorem. No portal is authorized.

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## 1 Claim boundary and field content

The candidate field list and branch convention are

$$\Psi_i \in \mathbf{5} \text{ Dirac}, \quad i = 1, 2; \quad \Phi_I \in \mathbf{4} \text{ complex}, \quad I = 1, 2; \quad \Sigma \in \mathbf{10} \text{ real.} \quad (1)$$

With the Hermitian generator  $H = -2iT_H^3$ ,

$$\mathbf{5} \longrightarrow \mathbf{2}_{+1} \oplus \mathbf{2}_{-1} \oplus \mathbf{1}_0, \quad \mathbf{4} \longrightarrow \mathbf{2}_0 \oplus \mathbf{1}_{+1} \oplus \mathbf{1}_{-1}. \quad (2)$$

The unbroken group is the direct product  $SU(2)_c \times U(1)_H$ , not its diagonal quotient. These facts use the injective  $Spin(4) \subset Spin(5)$  embedding and the standard low representations [1].

Four statements are separated throughout.

- (i) A common *quadratic* APS/PV regulator can be an exact elliptic construction around a nondegenerate smooth background.
- (ii) Such a construction is not a rigorous nonperturbative measure for an interacting four-dimensional gauge theory.
- (iii) A pure unbroken-gauge/gravitational mass-sign phase is a determinant ratio on bundles over spacetime.

- (iv) A target torsion phase additionally requires a map from target fields to a regulated Fredholm family and its determinant line.

The last distinction is the determinant-line version of the APS and Dai–Freed framework [2, 3, 4]. Recent bordism analyses of  $Sp$  QCD and mod-two  $Sp(n)$  anomalies support the same separation between local polynomials and torsion characters [5, 6].

## 2 Euclidean Hilbert spaces, gamma matrices, and reality

### 2.1 Geometric data and domains

Let  $M_4$  be a closed connected Riemannian spin four-manifold, let  $S = S^+ \oplus S^-$  be its complex spinor bundle, and let  $P \rightarrow M_4$  be a principal  $Sp(2)$ -bundle. Write  $E_R = P \times_R R$ . All connections are smooth and anti-Hermitian. The basic Hilbert spaces are

$$\mathcal{H}_F = L^2(M_4, S \otimes E_5 \otimes \mathbb{C}_f^2), \quad \text{Dom } \not{D}_A = H^1(M_4, S \otimes E_5 \otimes \mathbb{C}_f^2), \quad (3)$$

$$\mathcal{H}_B = L^2(M_4, \mathcal{B}), \quad \text{Dom } \Delta_B = H^2(M_4, \mathcal{B}), \quad (4)$$

$$\mathcal{H}_{\text{gh}} = L^2(M_4, \text{ad } P \otimes \mathbb{C}), \quad \text{Dom } \Delta_{\text{gh}} = H^2(M_4, \text{ad } P \otimes \mathbb{C}), \quad (5)$$

where

$$\mathcal{B} = (T^*M_4 \otimes \text{ad } P) \oplus \text{ad } P \oplus (E_4^{\oplus 2})_{\mathbb{R}}. \quad (6)$$

The last summand is realified because the scalar Gaussian is a real bosonic Gaussian. Compactness and ellipticity give discrete spectra with finite multiplicity. A prime on a determinant will remove the finite-dimensional kernel and place its orientation in the determinant line.

### 2.2 Clifford and real structures

Choose Hermitian Euclidean gamma matrices

$$\gamma_1 = \sigma_1 \otimes \sigma_1, \quad \gamma_2 = \sigma_1 \otimes \sigma_2, \quad \gamma_3 = \sigma_1 \otimes \sigma_3, \quad \gamma_4 = \sigma_2 \otimes \mathbf{1}, \quad \gamma_5 = \gamma_1 \gamma_2 \gamma_3 \gamma_4. \quad (7)$$

Thus

$$\gamma_\mu^\dagger = \gamma_\mu, \quad \{\gamma_\mu, \gamma_\nu\} = 2\delta_{\mu\nu}, \quad \gamma_5^\dagger = \gamma_5, \quad \gamma_5^2 = 1. \quad (8)$$

Put  $B = \gamma_2 \gamma_4$  and let  $K$  denote componentwise complex conjugation. Direct multiplication gives

$$B\gamma_\mu^* B^{-1} = \gamma_\mu, \quad BB^* = -\mathbf{1}. \quad (9)$$

The  $\mathbf{5}$  is the real vector representation of  $Spin(5)$ , so its bundle has a parallel conjugation  $K_{\mathbf{5}}^2 = +1$ . Therefore

$$J = (BK) \otimes K_{\mathbf{5}}, \quad J^2 = -1 \quad (10)$$

is the quaternionic real structure on the one-flavour spinor bundle. It commutes with the covariant Dirac operator and a real scalar reference mass. It does *not* commute with a fixed nonzero adjoint Higgs mass in the convention below. Indeed, in a real vector basis the anti-Hermitian generators  $T_{\mathbf{5}}^a$  are real antisymmetric, and hence

$$K_{\mathbf{5}} \widehat{\Sigma}_{\mathbf{5}} K_{\mathbf{5}}^{-1} = -\widehat{\Sigma}_{\mathbf{5}}, \quad JK_{A,\Sigma} J^{-1} = K_{A,-\Sigma}. \quad (11)$$

Thus  $J^2 = -1$  gives Kramers pairing only in the scalar-reference sector (or for a family involution that also sends  $\Sigma \mapsto -\Sigma$ ), not at a fixed generic Higgs background. Positivity below instead uses the elementary two-flavour determinant square and gamma-five Hermiticity; no fixed-adjoint-mass Kramers claim is made.

For anti-Hermitian  $A$ ,  $\not{D}_A^\dagger = -\not{D}_A$ . Define

$$\widehat{\Sigma}_{\mathbf{5}} = -i\Sigma^a T_{\mathbf{5}}^a, \quad \mathcal{K}_{A,\Sigma} = \not{D}_A + m + y\widehat{\Sigma}_{\mathbf{5}}, \quad H_{A,\Sigma} = \gamma_5 \mathcal{K}_{A,\Sigma}. \quad (12)$$

Then

$$\mathcal{K}^\dagger = \gamma_5 \mathcal{K} \gamma_5, \quad H^\dagger = H, \quad \text{Dom } H = H^1. \quad (13)$$

The light mass is kept at  $M_+ = \mu > 0$  during every determinant calculation; the limit  $\mu \downarrow 0$  is taken only after zero modes have been separated.

### 3 The full background-field quadratic complex

#### 3.1 Gauge action and gauge fixing

Fix a smooth background  $(\bar{A}, \bar{\Sigma}, \bar{\Phi})$  and write the fluctuation as  $b = (a, \sigma, \varphi) \in \Gamma(\mathcal{B})$ . At vanishing background fermions the infinitesimal gauge-action operator is

$$\mathcal{R}_{\bar{X}}\epsilon = (D_{\bar{A}}\epsilon, [\bar{\Sigma}, \epsilon], -\epsilon\bar{\Phi}), \quad \mathcal{R}_{\bar{X}} : H^1(\text{ad } P) \longrightarrow L^2(\mathcal{B}). \quad (14)$$

The notation  $-\epsilon\bar{\Phi}$  includes both fundamental copies and realification. The background Feynman gauge functional is the exact adjoint

$$\mathcal{F}_{\bar{X}}(b) = \mathcal{R}_{\bar{X}}^\dagger b, \quad S_{\text{gf}}^{(2)} = \frac{1}{2} \|\mathcal{F}_{\bar{X}}(b)\|^2. \quad (15)$$

Let  $S_{\text{bos}}$  be the AP-E6 Yang–Mills plus covariant kinetic and scalar-potential action. Its gauge-fixed Hessian and ghost operator are defined without dropping mixing terms by

$$\Delta_B = S_{\text{bos}}^{(2)}[\bar{X}] + \mathcal{R}_{\bar{X}}\mathcal{R}_{\bar{X}}^\dagger, \quad (16)$$

$$\Delta_{\text{gh}} = \mathcal{R}_{\bar{X}}^\dagger \mathcal{R}_{\bar{X}}. \quad (17)$$

This definition contains the gauge–adjoint–fundamental mixing and the full matrix of second derivatives of the declared potential. Their principal symbols are

$$\sigma_2(\Delta_B)(x, \xi) = |\xi|^2 \mathbf{1}_B, \quad \sigma_2(\Delta_{\text{gh}})(x, \xi) = |\xi|^2 \mathbf{1}_{\text{ad } P}. \quad (18)$$

Hence both are Laplace type and essentially self-adjoint on their stated Sobolev domains. Background stabilizers are projected from both operators. If  $\Delta_B$  has negative modes, the Gaussian contour is the steepest descent contour fixed by the upper lateral Agmon cut  $\theta_B = \pi - \varepsilon$ ,  $0 < \varepsilon \ll 1$ ; such a saddle is not called stable. This convention defines the quadratic determinant but does not turn a negative mode into a physical vacuum.

#### 3.2 What is and is not included

Equations (16)–(17) are preferable to listing only diagonal blocks: the latter would silently drop the Higgs-mechanism mixing. The quadratic functional is

$$S^{(2)} = \frac{1}{2} \langle b, \Delta_B b \rangle + \langle \bar{c}, \Delta_{\text{gh}} c \rangle + \sum_{i=1}^2 \langle \bar{\Psi}_i, \mathcal{K} \Psi_i \rangle. \quad (19)$$

This is the entire gauge–scalar–ghost–fermion quadratic complex about the declared background. Fermion-loop contributions to an effective bosonic Hessian arise after integrating Eq. (19); they are not part of the bare Gaussian operator and are not claimed here.

### 4 One common Pauli–Villars prescription

#### 4.1 Superdeterminant and regulator statistics

Use the four-node finite difference

$$c_a = (1, -3, 3, -1), \quad \Lambda_a^2 = a\Lambda^2, \quad a = 0, 1, 2, 3. \quad (20)$$

It obeys

$$\sum_a c_a a^r = 0 \quad (r = 0, 1, 2), \quad \sum_a c_a a^3 = -6. \quad (21)$$

For any positive Laplace-type eigenvalue  $\lambda$ ,

$$\sum_a c_a \log(\lambda + a\Lambda^2) = -\frac{2\Lambda^6}{\lambda^3} + O(\lambda^{-4}), \quad (22)$$

so the first three ultraviolet moments cancel. Define

$$\begin{aligned} Z_{\text{quad}}^{\text{PV}}(\Lambda) &:= \tau_F \prod_{a=0}^3 \det'_{\theta_B}(\Delta_B + \Lambda_a^2)^{-c_a/2} \det'(\Delta_{\text{gh}} + \Lambda_a^2)^{c_a} \\ &\quad \times \det'(H^2 + \Lambda_a^2)^{N_f c_a/2}, \quad N_f = 2. \end{aligned} \quad (23)$$

Here  $\tau_F$  is the fermion determinant-line phase defined in the next section. The last determinant is only its positive amplitude. It can be implemented with an auxiliary Pauli doublet

$$\hat{H}_a = H \otimes \sigma_3 + \Lambda_a \mathbf{1} \otimes \sigma_1, \quad \hat{H}_a^2 = H^2 + \Lambda_a^2, \quad (24)$$

whose spectrum is phase-paired. Thus amplitude subtraction cannot silently choose a torsion phase.

Equation (23) is first of all a zeta-determinant prescription. Integer powers in complex sectors admit the usual commuting or Berezin auxiliary-field representation, with multiplicity  $|c_a|$ . For the real bosonic sector the half-integer power is instead the spectral-cut square root of the determinant. We do not claim that a positive half-integer power of a symmetric second-order operator is a literal real-Grassmann Gaussian: such a local realization would require an additional antisymmetric first-order Pfaffian construction, which is not supplied here. The three elliptic determinant families are:

| sector                  | operator at node $a$              | determinant exponent |
|-------------------------|-----------------------------------|----------------------|
| real gauge plus scalars | $\Delta_B + a\Lambda^2$           | $-c_a/2$             |
| complex FP ghost        | $\Delta_{\text{gh}} + a\Lambda^2$ | $+c_a$               |
| two Dirac amplitudes    | $H^2 + a\Lambda^2$                | $N_f c_a/2 = c_a$    |

Every mass is a scalar endomorphism, so the regulator preserves background  $Sp(4)$  covariance. BRST covariance at quadratic order follows because the boson and ghost operators are built from the same  $\mathcal{R}_{\bar{X}}$ .

## 4.2 Finite local subtraction

The moment equations remove power terms but leave local logarithmic terms. Let  $A_4(\bar{X})$  be the total integrated Seeley–DeWitt coefficient of the supercomplex in Eq. (23). At a fixed scale  $\mu_R$  we define the finite scheme

$$\log Z_{\text{quad}}^{\text{ren}} = \text{FP}_{\Lambda \rightarrow \infty} \left[ \log Z_{\text{quad}}^{\text{PV}}(\Lambda) - A_4(\bar{X}) \log \frac{\Lambda^2}{\mu_R^2} - S_{\text{pow}}^{\text{loc}}(\Lambda; \bar{X}) \right]. \quad (25)$$

The finite local convention is fixed by

$$\theta_c = \theta_H = \theta_{\text{grav}} = 0 \quad \text{for positive reference masses,} \quad Z_{\text{quad}}^{\text{ren}}[0, VH, 0] > 0, \quad (26)$$

and by adding no separate target torsion invertible theory. This fixes all local de Rham counterterms of the quadratic reference problem. It cannot fix a pullback determinant line over a target space for which no pullback map has been specified.

**Proposition 4.1** (Quadratic regulator closure). *On a smooth compact background with projected stabilizer and a declared Agmon contour for every negative bosonic mode, Eqs. (3)–(26) define the full common quadratic fermion–PV–gauge–scalar–ghost amplitude. Together with the phase construction below,*

$$\text{common\_quadratic\_aps\_pv\_regulator\_specified=true}. \quad (27)$$

*This statement is deliberately weaker than a nonperturbative continuum measure.*

## 5 Five-dimensional interpolation, spectral cut, and APS domain

### 5.1 The common cylinder operator

Let  $Y_5 = M_4 \times [0, 1]$  with product geometry near both ends. Choose a smooth path of backgrounds  $X(t) = (A(t), \Sigma(t), \Phi(t))$ , constant near  $t = 0, 1$ , and a Hermitian fermion mass endomorphism  $\mathcal{M}(t)$ . The positive PV reference is

$$H_0 = \gamma_5(\not{D}_{A(0)} + M_*), \quad M_* > 0, \quad (28)$$

and the endpoint is

$$H_1 = \gamma_5(\not{D}_{A(1)} + m + y\hat{\Sigma}(1)). \quad (29)$$

The simplest mass interpolation is affine away from collars,

$$\mathcal{M}(t) = (1 - f(t))M_* + f(t)(m + y\hat{\Sigma}(t)), \quad f = 0 \text{ near } 0, \quad f = 1 \text{ near } 1. \quad (30)$$

No gap is assumed along the path; crossings are the information measured by spectral flow.

For the physical determinant-sign ratio we require gapped endpoints,

$$0 \notin \text{spec}(H_0) \cup \text{spec}(H_1), \quad (31)$$

and fix the physical spectral cut to  $\alpha = 0$ . Crossings through zero in the open cylinder remain allowed. For the separate mathematical statement about a Fredholm index, one may instead choose any real cut in the common endpoint resolvent,  $\alpha \notin \text{spec}(H_0) \cup \text{spec}(H_1)$ , and write  $P_I(H)$  for the spectral projector of  $H$  on  $I$ . The Fredholm operator

$$\mathcal{D}_5^+ = \partial_t + H(t) \quad (32)$$

has the explicit domain

$$\begin{aligned} \text{Dom}_{\text{APS}, \alpha}(\mathcal{D}_5^+) &= \{u \in H^1(Y_5) : P_{[\alpha, \infty)}(H_0)u|_{t=0} = 0, \\ &\quad P_{(-\infty, \alpha)}(H_1)u|_{t=1} = 0\}. \end{aligned} \quad (33)$$

Its adjoint is  $-\partial_t + H(t)$  with complementary boundary projectors. The self-adjoint double is

$$\mathbb{D}_{5, \alpha} = \begin{pmatrix} 0 & (\mathcal{D}_5^+)^* \\ \mathcal{D}_5^+ & 0 \end{pmatrix} \quad (34)$$

on the direct sum of these two APS domains. This gives a literal Hilbert space and domain, rather than a schematic “five-dimensional eta operator.”

### 5.2 Reduced eta and determinant phase

For a self-adjoint elliptic endpoint operator define

$$\eta_\alpha(H; s) = \sum_{\lambda \neq \alpha} \text{sgn}(\lambda - \alpha) |\lambda - \alpha|^{-s}, \quad \bar{\eta}_\alpha(H) = \frac{\eta_\alpha(H; 0) + \dim \ker(H - \alpha)}{2}. \quad (35)$$

For every such auxiliary cut, the APS theorem identifies

$$\text{ind}_{\text{APS}, \alpha}(\mathcal{D}_5^+) = \text{SF}_\alpha\{H(t)\}. \quad (36)$$

For a real mass interpolation with the gapped endpoints (31), the regulated physical determinant-sign ratio is

$$\tau_F[X(t)] = \exp(i\pi N_f \text{SF}_0\{H(t)\}). \quad (37)$$

The arbitrary  $\alpha$  in Eq. (36) is therefore an index bookkeeping device and must not be substituted into Eq. (37). A formulation that moves the physical cut would require the corresponding endpoint

spectral-section orientation (or reduced-eta correction); that extra datum is not used here. Equivalent zero-cut reduced-eta formulae differ by the local index density already fixed in Eq. (26). APS gluing and the Bismut–Freed holonomy theorem make the resulting zero-cut section compatible with concatenating paths [2, 3, 4].

For a general complex chiral mass family the phase is a section of the pulled-back determinant line, not simply a sign. The Dai–Freed prescription still applies, but only after that family and its endpoint bundles have been defined [7, 8].

## 6 A conditional GW-compatible Higgsed source

### 6.1 Finite compact-link regulator

There is a useful regulator framework that goes beyond the continuum Gaussian without pretending to solve the continuum limit or already supplying the target Yukawa operator. On a finite four-dimensional hypercubic lattice take compact  $Sp(4)$  links  $U_\ell$  with Haar measure, the Wilson plaquette action, and site fields  $\Sigma_x, \Phi_x$  with the gauge-covariant nearest-neighbour discretization of the declared scalar action. The finite bosonic integral is gauge invariant and requires no gauge fixing or Faddeev–Popov ghosts. Its perturbative expansion is matched to the continuum background complex of Section 3.

Let

$$H_W(U) = \gamma_5(D_W(U) - m_0), \quad 0 < m_0 < 2, \quad D_{\text{ov}} = \frac{1}{a}(1 + \gamma_5 \text{sgn } H_W). \quad (38)$$

At finite  $a$ , chiral sources use the Ginsparg–Wilson involution and projectors

$$\hat{\gamma}_5 = \gamma_5(1 - aD_{\text{ov}}), \quad \hat{P}_{L,R} = \frac{1 \mp \hat{\gamma}_5}{2}; \quad (39)$$

the un-hatted  $P_{L,R}$  in the continuum mass formula below denote their  $a \rightarrow 0$  limits. This keeps the finite-lattice source compatible with the overlap chiral decomposition. On the admissible-link locus,  $H_W$  is gapped and the sign function is local. A five-dimensional domain-wall determinant divided by its Pauli–Villars field of dimensionless mass  $m_{\text{PV}} = 1$  converges as  $L_s \rightarrow \infty$  to the overlap determinant. Domain-wall eta invariants realize the APS and mod-two APS indices, while the massive Wilson eta invariant agrees with the continuum index at sufficiently small lattice spacing [9, 10, 11]. These facts provide a finite regulator for a Dirac operator once that operator and its domain-wall interpolation have been specified. They do not by themselves define the target-dependent Dirac–Yukawa family used below.

### 6.2 Spectral projectors and an explicit target mass

Restrict now to the exact Higgs orbit on which

$$h = \frac{\hat{\Sigma}_5}{V}, \quad \text{spec } h = (+1, +1, -1, -1, 0), \quad h^3 = h. \quad (40)$$

The covariant orthogonal projectors are

$$P_+ = \frac{h^2 + h}{2}, \quad P_- = \frac{h^2 - h}{2}, \quad P_0 = 1 - h^2. \quad (41)$$

They obey

$$P_r P_s = \delta_{rs} P_r, \quad P_+ + P_- + P_0 = 1, \quad (\text{rank } P_+, \text{rank } P_-, \text{rank } P_0) = (2, 2, 1). \quad (42)$$

Away from the exact orbit but before a spectral gap closes, the same projectors are defined by Riesz contours around the three separated spectral clusters. They cease to exist as branch projectors at the restored locus.

Let  $g_x \in SU(2)_f$  be a declared Hubbard–Stratonovich chiral mass source. Let  $A_H(s)$  be the pullback Hopf connection for  $s : M_4 \rightarrow S^2$ , embedded into the links through  $\exp(iA_H H)$ . The continuum Higgsed all-branch source is

$$\begin{aligned}\mathcal{M}_H(g, s; \Sigma) &= P_+ \otimes \mu(P_L g + P_R g^\dagger) - 2yVP_- \otimes \mathbf{1}_f - yVP_0 \otimes \mathbf{1}_f, \\ P_L &= \frac{1 - \gamma_5}{2}, \quad P_R = \frac{1 + \gamma_5}{2}.\end{aligned}\tag{43}$$

Here  $s$  is intended to enter the covariant Wilson/overlap operator through its Hopf link, not as an undeclared direct Yukawa. Equations (38)–(43) specify a GW-compatible continuum-limit source on the gapped Higgsed locus. They do not specify the finite operator: the ordinary projectors acting on  $\bar{\Psi}$ , the hatted projectors acting on  $\Psi$ , determinant orientation, and the corresponding five-dimensional target kernel remain to be written. Consequently this note does not claim a completed overlap determinant line or a computed overlap eta invariant.

There is nevertheless a useful conditional copy-counting diagnostic. On  $G_3$ , the unit  $S^3$  mass winding is repeated rank  $P_+ = 2$  times, hence its mod-two index is zero. On  $G_2$ , the light charged sector has  $2 \cdot 2 \cdot |c_1| = 4$  Hopf-flux zero-mode copies; the charged heavy sector adds another four, and the neutral sector adds none. Therefore, assuming a finite GW implementation with the displayed continuum limit, standard copy-additivity and orientation, positive  $m_{PV} = 1$ , and no stacked  $I_3, I_2$  sector, the predicted pair is

$$(\tau_{H,ov}(G_3), \tau_{H,ov}(G_2)) = (+1, +1).\tag{44}$$

This is a conditional parity prediction, not an eta computation or a unique phase extracted from the original deep-UV action.

**Fail-closed gate 6.1** (Higgsed construction versus UV claim). The exact distinction is

$$\begin{aligned}\text{conditional\_higgsed\_gw\_source\_specified} &= \text{true}, \\ \text{finite\_higgsed\_overlap\_dirac\_yukawa\_operator\_defined} &= \text{false}, \\ \text{five\_dimensional\_higgsed\_target\_kernel\_defined} &= \text{false}, \\ \text{minimal\_higgsed\_overlap\_eta\_pair\_computed} &= \text{false}, \\ \text{conditional\_higgsed\_parity\_predicts\_plus\_plus} &= \text{true}, \\ \text{original\_deep\_sp4\_action\_uniquely\_selects\_target\_pair} &= \text{false}.\end{aligned}\tag{45}$$

## 7 The two target generators: what the common regulator computes

### 7.1 Bordism representatives and abstract characters

Let  $X = S^3 \times S^2$ . The two reduced spin-bordism generators are

$$G_3 = (S_R^1 \times S^3, \text{pr}_{S^3}, \text{pt}),\tag{46}$$

$$G_2 = (T_{RR}^2 \times S^2, \text{pt}, \text{pr}_{S^2}).\tag{47}$$

For regular values define

$$\nu_3(M, g) = \dim_{\mathbb{R}} \ker D_{g^{-1}(p_3)} \pmod{2}, \quad \nu_2(M, s) = \text{Arf}(s^{-1}(p_2)).\tag{48}$$

Then

$$(\nu_3(G_3), \nu_2(G_3)) = (1, 0), \quad (\nu_3(G_2), \nu_2(G_2)) = (0, 1),\tag{49}$$

because the periodic circle is nonbounding and the RR torus has odd spin structure [12]. The four abstract torsion characters are

$$Z_{\epsilon_3, \epsilon_2}(M, g, s) = (-1)^{\epsilon_3 \nu_3 + \epsilon_2 \nu_2}, \quad (\epsilon_3, \epsilon_2) \in \mathbb{Z}_2^2.\tag{50}$$

## 7.2 Pure heavy component on both representatives

The common regulator can be evaluated on the part of each representative that is already a declared unbroken gauge bundle. On  $G_3$ , take the unbroken bundle trivial. Its characteristic numbers are

$$\int c_2(V_c) = 0, \quad \int x^2 = 0, \quad \int p_1(TG_3) = 0. \quad (51)$$

On  $G_2$ , take  $L = \text{pr}_{S^2}^* \mathcal{O}(1)$ , so  $x = \text{pr}_{S^2}^* u$ . Since  $u^2 = 0$  on  $S^2$ , and the product tangent bundle has vanishing Pontryagin number,

$$\int c_2(V_c) = 0, \quad \int x^2 = 0, \quad \int p_1(TG_2) = 0. \quad (52)$$

The heavy-index formula proved below therefore gives

$$\boxed{\tau_h^{\text{pure}}(G_3) = +1, \quad \tau_h^{\text{pure}}(G_2) = +1.} \quad (53)$$

These are actual evaluations of the constant-mass unbroken-gauge plus gravitational determinant ratio. They do not contain a pion chiral mass or an  $s$ -dependent fermion Yukawa family.

## 7.3 No target pullback: a precise non-identifiability theorem

Let  $\mathcal{C}_{UV}$  be the quotient of declared smooth  $(A, \Sigma, \Phi)$  configurations by gauge transformations. The common quadratic regulator defines a determinant line  $\mathcal{L}_{UV}$  over the nondegenerate locus of  $\mathcal{C}_{UV}$ . To ask for a target eta pair one needs a map

$$\mathfrak{L} : \text{Map}(M_4, S^3 \times S^2) \longrightarrow \mathcal{C}_{UV}, \quad (g, s) \longmapsto (A, \Sigma, \Phi, \mathcal{M}_{\text{all branches}}), \quad (54)$$

and then the line  $\mathfrak{L}^* \mathcal{L}_{UV}$ . The declared microscopic action has no such map:  $g$  is a composite pion orientation, not an elementary mass source, and  $\Phi$  does not enter the fermion mass  $m + y\hat{\Sigma}_5$ . A Hopf connection associated with  $s$  can specify the  $U(1)_H$  bundle used in Eq. (52); it does not supply the missing chiral mass on all light and heavy branches. The continuum source in Eq. (43) does not yet supply this map: a finite GW Dirac–Yukawa operator, its five-dimensional kernel, and determinant orientation are still absent. Even after those are added, the resulting map would depend on a new Hubbard–Stratonovich source, a no-stack convention, and a restriction to the gapped Higgs orbit, rather than being data implicit in the original action.

**Theorem 7.1** (Target eta non-identifiability). *The data in Eqs. (1)–(26) determine the pure pair in Eq. (53), but do not determine  $(\tau_{\text{full}}(G_3), \tau_{\text{full}}(G_2))$ . In particular, setting “no torsion stack” in the UV local scheme cannot replace the absent map  $\mathfrak{L}$ .*

*Proof.* Without  $\mathfrak{L}$  in the original action, the pullback determinant line on which the two holonomies would be evaluated is undefined. This is a type obstruction, not a difficult numerical eta sum. Independently, suppose one arbitrary finite lift with continuum limit Eq. (43) were supplied. Stacking either invertible spin character  $I_3 = (-1)^{\nu_3}$  or  $I_2 = (-1)^{\nu_2}$  has zero local de Rham curvature, changes none of the principal symbols, PV moments, gauge-bordism character, or pure heavy index, but flips exactly one target value. Thus all four rows

$$(+, +), \quad (-, +), \quad (+, -), \quad (-, -)$$

remain compatible with the currently declared local and pure-threshold data. Selection requires both  $\mathfrak{L}$  and an explicit microscopic choice of the torsion invertible sector.  $\square$

For comparison, the circle spectral path  $D_{S^1, \alpha_s} + t$ ,  $-1/4 \leq t \leq 1/4$ , has one crossing for periodic spin  $\alpha_s = 0$  and none for antiperiodic spin  $\alpha_s = 1/2$ . The RR torus has one chiral zero mode modulo two, whereas the other three torus spin structures have none. These exact KO controls reproduce Eq. (49); they realize the two abstract stacks, not the missing  $Sp(4)$  target lift.

**Fail-closed gate 7.2** (Full target pair). The exact status is

$$\begin{aligned}
&\text{pure\_heavy\_phase\_on\_G3\_computed}=\text{true}, \\
&\text{pure\_heavy\_phase\_on\_G2\_computed}=\text{true}, \\
&\text{conditional\_higgsed\_parity\_predicts\_plus\_plus}=\text{true}, \\
&\text{minimal\_higgsed\_overlap\_eta\_pair\_computed}=\text{false}, \\
&\text{original\_action\_target\_mass\_family\_lift\_defined}=\text{false}, \\
&\text{target\_mapping\_torus\_eta\_pair\_selected}=\text{false}.
\end{aligned} \tag{55}$$

## 8 Exact constant-mass heavy threshold

### 8.1 Index polynomial

At  $m = -yV + \mu$ , with  $yV > 0$  and then  $\mu \downarrow 0$ , the branches have signs

| branch        | bundle               | charge | limiting mass | sign |
|---------------|----------------------|--------|---------------|------|
| light         | $V_c \otimes L$      | +1     | $\mu$         | +    |
| charged heavy | $V_c \otimes L^{-1}$ | -1     | $-2yV$        | -    |
| neutral heavy | $\mathbf{1}$         | 0      | $-yV$         | -    |

Put  $x = c_1(L)$  and use  $c_1(V_c) = 0$ . The degree-four Chern characters are

$$\text{ch}_2(V_c \otimes L^{-1}) = -c_2(V_c) + x^2, \tag{56}$$

$$\text{ch}_2(\mathbf{1}) = 0. \tag{57}$$

Since  $\widehat{A}(TM) = 1 - p_1(TM)/24 + \dots$ , the one-flavour heavy index is

$$\begin{aligned}
I_h &= \int_{M_4} \widehat{A}(TM_4) [\text{ch}(V_c \otimes L^{-1}) + 1]_{(4)} \\
&= \int_{M_4} \left[ -c_2(V_c) + x^2 - \frac{1}{8}p_1(TM_4) \right] \in \mathbb{Z}.
\end{aligned} \tag{58}$$

The separate gravitational coefficients are  $-p_1/12$  for the rank-two charged branch and  $-p_1/24$  for the singlet. Thus Eq. (58) includes both and does not discard the neutral fermion.

For one negative Dirac mass relative to a positive reference, paired nonzero eigenvalues cancel and the zero modes give

$$\frac{\det(\not{D} - |M|)}{\det(\not{D} + |M|)} = (-1)^{\dim \ker \not{D}} = (-1)^{\text{ind } D^+}. \tag{59}$$

The second equality holds because sum and difference of the two chiral kernel dimensions have the same parity. It is also the spectral-flow evaluation of Eq. (37).

**Theorem 8.1** (Two-flavour pure threshold). *For the complete negative-mass heavy sector and  $N_f = 2$ ,*

$$\tau_h^{\text{pure}}(M_4, V_c, L) = \exp(i\pi N_f I_h) = \exp(2\pi i I_h) = 1 \tag{60}$$

*on every closed spin four-manifold and every  $SU(2)_c \times U(1)_H$  bundle.*

*Proof.* Equation (59) applies additively to the charged and neutral heavy branches, giving  $I_h$  in Eq. (58). The index is integral by the Atiyah–Singer theorem. Two identical flavours multiply it by two, so the exponential is one. Equivalently, on a closed spin four-manifold  $x^2$  is even and  $p_1 = 3\sigma$  with  $\sigma \in 16\mathbb{Z}$ , while the possible odd  $c_2(V_c)$  contribution is also doubled.  $\square$

The corresponding finite local threshold functional is

$$S_{h,\text{loc}} = 2\pi i \int_{M_4} \left[ -c_2(V_c) + x^2 - \frac{1}{8}p_1(TM_4) \right]. \quad (61)$$

It is an integral counterterm and exponentiates to one on closed spin backgrounds. On a manifold with boundary its Chern–Simons transgression is precisely the local term paired with the APS eta section; the finite scheme in Eq. (26) removes it. Thus the constant real-mass unbroken-gauge and gravitational determinant ratio is actually matched, not merely checked for theta periodicity.

**Fail-closed gate 8.2** (Scope of the threshold theorem). Equation (60) closes

$$\text{pure\_unbroken\_gauge\_gravity\_heavy\_phase\_computed}=\text{true}. \quad (62)$$

It does not evaluate a target-dependent chiral mass family, because that family is absent from the action. Accordingly

$$\text{mixed\_wzw\_heavy\_determinant\_computed}=\text{false}. \quad (63)$$

## 9 Controlled all-scale mixed-WZW tests

The mixed WZW curvature studied in the recent UV completion of Davighi–Lohitsiri is an integral degree-five class and its coefficient is fixed by a generalized-symmetry matching condition [13]. For the present branch weights, a uniform exact chiral flavour action on the complete **5** gives

$$\kappa_{\text{full}} = 2(+1) + 2(-1) + 1(0) = 0, \quad \kappa_{\text{light}} = +2, \quad \kappa_{\text{heavy}} = -2. \quad (64)$$

The sign of a real Dirac mass does not change this charge trace: multiplying an  $SU(2)$  chiral mass map by  $-1$  leaves  $(-g)^{-1}d(-g) = g^{-1}dg$ .

**Theorem 9.1** (Exact separated rank-two projector obstruction). *Let  $\mathcal{U}$  be an  $Sp(4)$ -invariant neighbourhood of  $\Sigma = 0$  in the adjoint representation. Suppose*

$$C : \mathcal{U} \longrightarrow \text{End}_{\mathbb{C}}(\mathbf{5})$$

*is continuous and equivariant,  $C(u\Sigma) = R_{\mathbf{5}}(u)C(\Sigma)R_{\mathbf{5}}(u)^{-1}$ . If  $C(t\Sigma_0)$  is the rank-two light-branch projector  $P_+$  for  $t > 0$ , then  $C$  cannot extend to  $t = 0$  while retaining a separated rank-two branch. More generally, a continuous equivariant chiral-mass coefficient has*

$$C(0) = c\mathbf{1}_{\mathbf{5}}, \quad \text{Tr}_{\mathbf{5}}(HC(0)) = c \text{Tr}_{\mathbf{5}} H = 0, \quad (65)$$

*whereas the Higgsed selector obeys*

$$\text{Tr}_{\mathbf{5}}(HP_+) = 2. \quad (66)$$

*Proof.* The restored point is fixed by all of  $Sp(4)$ . Equivariance therefore makes  $C(0)$  commute with the irreducible complex **5**; Schur’s lemma gives the first equation in Eq. (65). The eigenvalues of  $H = h(\Sigma_0)$  are  $+1, +1, -1, -1, 0$ , so its trace is zero, while Eq. (41) gives  $\text{Tr}(HP_+) = 1 + 1 = 2$ . Continuity along  $t\Sigma_0$  is incompatible with keeping this coefficient equal to two at  $t = 0$ . Equivalently, if  $C$  remains an orthogonal projector, then  $C(0) = c\mathbf{1}$  and  $C(0)^2 = C(0)$  force  $c = 0$  or  $1$ , of rank zero or five, never rank two. Hence the branch gap must close, the selector must become discontinuous, equivariance must fail, or new UV fields must carry the missing data.  $\square$

This theorem is deliberately limited to an exact separated projector. It does not exclude continuous equivariant non-idempotent coefficients such as  $C(\Sigma) = c\mathbf{1} + f(\|\Sigma\|)\widehat{\Sigma}$ . Such families can reorganize their spectrum or close a gap at the restored point and require a separate determinant-line analysis.

**Proposition 9.2** (Two controlled mass assignments). *Within the currently declared  $Sp(4)$  field/action data, the following two assignments can be evaluated:*

- (A) the chiral flavour mass map acts uniformly on the full **5**, in which case the exact all-scale mixed coefficient is zero by Eq. (64); or
- (B) the desired chiral flavour symmetry emerges only on the tuned light branch below the breaking threshold, in which case no deep-UV target mass-family lift exists and its APS eta pair cannot be computed.

Neither evaluated assignment is an all-scale derivation of  $k = +2$ . No claim is made that these two assignments exhaust all continuous equivariant non-projector mass families.

*Proof.* In case (A), additivity of the family index over representation weights gives the first equality in Eq. (64); local four-dimensional counterterms cannot change the integral cohomology class of the degree-five curvature. Restoring  $+2$  requires adding a new heavy differential character or changing the representation/symmetry assignment. In case (B), the heavy Dirac masses preserve only vector flavour, so the deep-UV operator is not a family over  $S^3$ ; this is exactly the type obstruction in Eq. (54). New fields, a new invertible sector, or a non-projector equivariant mass family require additional analysis outside these two controlled assignments.  $\square$

The infrared orientation remains meaningful:

$$n_{\text{IR}} = N_c X_q = 2, \quad k_{\text{IR}} = n_{\text{IR}} B = +2 \quad (X_q = +1, B = +1). \quad (67)$$

It is not promoted to an all-scale theorem. Recent  $Sp$ -QCD bordism work controls the consistency of WZW sectors but does not provide the missing mixed target family of this particular Higgsed model [5].

## 10 Deterministic verification

The companion program `route_f/code/verify_ap_e7_common_sp4_aps_pv_regulator.py` performs exact or floating-point checks of:

1. the Euclidean Clifford algebra, gamma-five, quaternionic reference structure  $BB^* = -1$ , and the odd transformation of the adjoint mass;
2. gamma-five Hermiticity and the two-flavour determinant square for a scalar-reference control;
3. all PV moments and the  $-2/\lambda^3$  asymptotic coefficient;
4. positivity and common construction of toy boson and ghost Laplace symbols;
5. the physical zero-cut sign, an auxiliary-cut counter-control, R/NS circle spectral flow, and the four abstract target characters;
6. a symbolic Chern-character expansion, spin-integrality spot checks on a finite characteristic-number grid, and direct evaluation of the two pure generator indices;
7. the conditionally derived Higgsed parity prediction, the  $+2 - 2 = 0$  charge trace, and explicitly labelled declared-scope gates.

These checks verify the algebra and reproducibility. They are not a lattice definition of interacting  $Sp(4)$  gauge theory. This regulator branch is background-field/topological and does not import or depend on the relaxed  $B = 1$  soliton profile or its checksum.

## 11 Final gate ledger and ordered continuation

| claim                                                         | status                    | exact boundary                                                                                                                                                   |
|---------------------------------------------------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Euclidean gamma and reference-reality data                    | closed with scope         | explicit Clifford representation, $J^2 = -1$ , scalar-reference Kramers sector; fixed adjoint mass transforms with $\Sigma \mapsto -\Sigma$                      |
| Full common quadratic gauge–scalar–ghost–fermion PV amplitude | closed                    | exact gauge complex, all mixing retained, shared four-node superdeterminant and finite local scheme                                                              |
| Five-dimensional APS problem                                  | closed for UV field paths | zero-cut physical phase with gapped endpoints; arbitrary cut retained only for the Fredholm index; complementary APS domains and self-adjoint double             |
| Nonperturbative continuum $Sp(4)$ measure                     | open                      | no constructive interacting cutoff removal, global gauge slice, or reflection-positivity proof                                                                   |
| Pure unbroken gauge plus gravitational heavy threshold        | closed                    | two-flavour phase $\exp(2\pi i I_h) = 1$ , local integral counterterm explicit                                                                                   |
| Pure heavy phase on $G_3, G_2$                                | closed                    | $(+1, +1)$ for the constant-mass unbroken bundle component                                                                                                       |
| Conditional Higgsed GW source                                 | source only               | $(+1, +1)$ is a copy-counting prediction; finite Dirac–Yukawa operator, five-dimensional target kernel, determinant orientation, and eta computation remain open |
| Full target eta pair                                          | undefined                 | no all-branch target mass map or pullback determinant line in the original action; two torsion stacks prove non-identifiability                                  |
| Uniform exact-flavour assignment                              | evaluated, gives zero     | complete-irrep coefficient $+2 - 2 = 0$ ; not an exhaustive classification of equivariant non-projector families                                                 |
| Light emergent-flavour $k_{\text{IR}} = +2$                   | derived only in IR        | no deep-UV target-family APS matching                                                                                                                            |
| Degree-one Route-E portal                                     | forbidden                 | neither target matching nor full $Sp(4)$ dynamical lane is closed                                                                                                |

The machine-readable final flags are

$$\begin{aligned}
&\text{euclidean\_gamma\_reference\_reality\_data\_specified}=\text{true}, \\
&\text{exact\_separated\_rank\_two\_projector\_obstruction\_proven}=\text{true}, \\
&\quad \text{common\_quadratic\_aps\_pv\_regulator\_specified}=\text{true}, \\
&\quad \text{five\_dimensional\_aps\_domain\_specified}=\text{true}, \\
&\quad \text{physical\_fermion\_phase\_zero\_cut\_fixed}=\text{true}, \\
&\quad \text{fixed\_adjoint\_mass\_kramers\_reality\_proven}=\text{false}, \\
&\quad \text{full\_nonperturbative\_sp4\_measure\_constructed}=\text{false}, \\
&\quad \text{pure\_unbroken\_gauge\_gravity\_heavy\_phase\_computed}=\text{true}, \\
&\quad \text{conditional\_higgsed\_gw\_source\_specified}=\text{true}, \\
&\text{finite\_higgsed\_overlap\_dirac\_yukawa\_operator\_defined}=\text{false}, \\
&\quad \text{five\_dimensional\_higgsed\_target\_kernel\_defined}=\text{false}, \\
&\quad \text{minimal\_higgsed\_overlap\_eta\_pair\_computed}=\text{false}, \\
&\quad \text{conditional\_higgsed\_parity\_predicts\_plus\_plus}=\text{true}, \\
&\quad \text{original\_action\_target\_mass\_family\_lift\_defined}=\text{false}, \\
&\quad \text{target\_mapping\_torus\_eta\_pair\_selected}=\text{false}, \\
&\quad \text{mixed\_wzw\_heavy\_determinant\_computed}=\text{false}, \\
&\quad \text{k\_plus\_two\_all\_scale\_matched}=\text{false}, \\
&\quad \text{sp4\_aps\_pv\_lane\_closed}=\text{false}, \\
&\quad \text{degree\_one\_route\_e\_portal\_started}=\text{false}, \\
&\quad \text{physics\_promotion\_authorized}=\text{false}.
\end{aligned} \tag{68}$$

The next discriminating calculations are:

1. write the exact finite GW Dirac–Yukawa operator, its determinant orientation, and its five-dimensional domain-wall target kernel before promoting the conditional (+1, +1) parity count to an eta result;
2. derive the Hubbard–Stratonovich source in Eq. (43) from the original interacting action, or replace it by new UV fields that evade the exact-projector obstruction;
3. if exact uniform flavour is retained, add a specified heavy differential character of coefficient +2 and calculate both of its torsion signs rather than choosing them;
4. alternatively, formulate a symmetry-emergence matching theorem across the  $Sp(4)$  breaking scale, including the operator that creates the light-only chiral determinant line;
5. only after one of these closes, combine it with the independent relaxed-soliton or  $SO(3)/FR$  determinant-line routes. The portal remains last.

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